

Joao Coelho

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Summary

Joao Coelho is a PhD candidate in a Dual-PhD program between Carnegie Mellon University and Instituto Superior Técnico, expecting to graduate mid 2027. His current research focuses on dense retrieval and deep search systems, with broader interests in Machine Learning, Natural Language Processing, and Information Retrieval.

Selected Publications

Preprints

- Joao Coelho, Joao Magalhaes, Bruno Martins, Chenyan Xiong. **Effective Reinforcement Learning for Agentic Search by Recycling Zero-Variance Queries During Training**. ArXiv (under review), 2026.
- Sidhaarth Murali, Joao Coelho, Jingjie Ning, Joao Magalhaes, Bruno Martins, Chenyan Xiong. **Beyond Parallel Sampling: Diverse Query Initialization for Agentic Search**. ArXiv (under review), 2026.

International Conferences

- Joao Coelho, Jingjie Ning, Jingyuan He, Kangrui Mao, Abhijay Sai Paladugu, Pranav Setlur, Jiahe Jin, Jamie Callan, Joao Magalhaes, Bruno Martins, Chenyan Xiong. **DeepResearchGym: A Free, Transparent, and Reproducible Sandbox for Deep Research**. Proceedings of the ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval, 2026.
- Jingjie Ning*, Joao Coelho*, Yibo Kong*, Yunfan Long, Bruno Martins, Joao Magalhaes, Jamie Callan, Chenyan Xiong. **Agentic Search in the Wild: Intents and Trajectory Dynamics from 14M+ Real Search Requests**. Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval, 2026. (*equal contribution)
- Joao Coelho, Bruno Martins, Joao Magalhaes, and Chenyan Xiong. **Aligning Web Query Generation with Ranking Objectives via Direct Preference Optimization**. Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval, 2025.
- Joao Coelho, Bruno Martins, Joao Magalhaes, Jamie Callan, and Chenyan Xiong. **Dwell in the Beginning: How Language Models Embed Long Documents for Dense Retrieval**. Proceedings of the Annual Meeting of the Association for Computational Linguistics, 2024.
- Joao Coelho, Joao Magalhaes, and Bruno Martins. **Improving Neural Models for the Retrieval of Relevant Passages to Geographical Queries**. Proceedings of the ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems, 2021.

International Journals

- Joao Coelho, Beatriz Paula, Diogo Mano, Carlos Coutinho, Joao Oliveira, Ricardo Ribeiro, and Fernando Batista. **Semantic Similarity for Mobile Application Recommendation under Scarce User Data**. Engineering Applications of Artificial Intelligence, 2023.

Experience

Applied Scientist Intern, Amazon Web Services – Seattle, WA May 2025 – Aug 2025

- Built distributed training and inference infrastructure to support extensive dense and sparse retrieval experimentation across knowledge bases scaling up to 136 million documents.
- Proposed a search-aware multi-turn query rewriting agent, which achieved 10% absolute hit rate improvement over the team's production solution on internal customer service search benchmarks.
- Improved efficiency by distilling the agentic behavior into a single-turn locally deployable language model, retaining 98% of the performance while running 4 times faster.

Machine Learning Engineer, Caixa Mágica Software – Lisbon, Portugal Sep 2020 – Sep 2022

- Proposed self-supervised BERT-based recommendation system for mobile applications achieving 10% absolute improvement in real-user satisfaction from A/B testing on a major app store.

- Led and presented a proof-of-concept integrating graph databases and neural entity extraction for the Portuguese Archive, securing one of the company’s largest client projects.
- Built the machine learning pipeline that extended this proof-of-concept, enabling entity extraction and disambiguation over a large historical document collection.
- Supervised interns across recommendation and NLP projects; led technical workshops and co-authored multiple peer-reviewed papers.

Intern Developer, Caixa Mágica Software – Lisbon, Portugal Jul 2020 – Sep 2020

- Built FAQ retrieval system for automated helpdesk platform under low data resources.
- Deployed the solution by integrating retrieval models with a React front-end using Docker and CI/CD pipelines.

Intern Developer, Xpand IT – Lisbon, Portugal Jul 2019 – Sep 2019

- Implemented proactive chatbot with notification and polling capabilities for internal MS Teams integration.
- Integrated LUIS language understanding and deployed production system using .NET and Azure infrastructure.

Teaching

Teaching Assistant, Carnegie Mellon University – Pittsburgh, PA

- Large Language Models: Methods and Applications (Fall 2024)

Teaching Assistant, Instituto Superior Técnico – Lisbon, Portugal

- Search and Planning (Fall 2021, Fall 2022)
- Artificial Intelligence (Spring 2021, Spring 2022)

Education

Carnegie Mellon University, Pittsburgh, PA – Dual PhD in Language Technologies Sep 2022 - Present

Instituto Superior Técnico, Lisbon, Portugal – Dual PhD in Electrical Engineering Sep 2022 - Present

Instituto Superior Técnico, Lisbon, Portugal – MSc in Computer Science and Engineering Sep 2021

- Merit Distinctions: 2019/2020 and 2020/2021.
- Thesis: Geographical Question Answering Leveraging Neural Language Models for Passage Retrieval

Instituto Superior Técnico, Lisbon, Portugal – BSc in Computer Science and Engineering Sep 2019

- Merit Distinctions: 2017/2018 and 2018/2019.

Skills

Machine Learning & NLP: PyTorch, Transformers, LLM (pre-, mid-, post-)training, Knowledge Distillation, Dense Retrieval, Re-ranking, ANN (FAISS), Recommendation Systems, Question Answering, RAG.

Engineering & Cloud: Distributed Training/Inference, AWS, Azure, Docker, CI/CD.

Databases: SQL, NoSQL, Graph Databases (Neo4J), Elasticsearch

Languages: Portuguese (native), English (C1).